The logarithm and the logarithmic function

The inverse of the exponential — a new name for an index, and the graph you get by reflecting.

LESSON 57–58 | 2 × 40 MIN | ALGEBRA 10, §3.4 | P2 · 2.2

LESSON CLOCK

12' | 20' | 24' | 20' | 18' | 6'

The definition	12 min
Evaluating	20 min
The graph	24 min
Inverse of the exponential	20 min
Practice	18 min
Homework	6 min

LEARNING OBJECTIVES

- State the definition of a logarithm and move between the two forms.
- Evaluate simple logarithms without a calculator.
- Sketch y = log_a x and state its domain, range and asymptote.
- Recognise the logarithm as the inverse of the exponential.

TEXTBOOK

National	pp. 225–236
Cambridge	Pure Mathematics 2 & 3, pp. 15–25
Workbook	P2 Exercise 2D

01 Explanation

The definition

A LOGARITHM *IS* AN INDEX

$$log\_a\ b = c \Leftrightarrow a^\wedge c = b\ (a > 0, a \neq 1, b > 0)$$

In words: ***log_a b* is the power to which *a* must be raised to give *b*.**

So $log_2 8 = 3$ because $2^3 = 8$; $log_5 25 = 2$ because $5^2 = 25$; and $log_3 \frac{1}{9} = -2$ because $3^\wedge(-2) = \frac{1}{9}$.

THE ARGUMENT MUST BE POSITIVE

$log\_a(-4)$ does not exist: no power of a positive base is negative. That single condition is the first line of every logarithmic equation in the next four lessons.

Evaluating without a calculator

Ask “a to what power gives b?” and answer it.

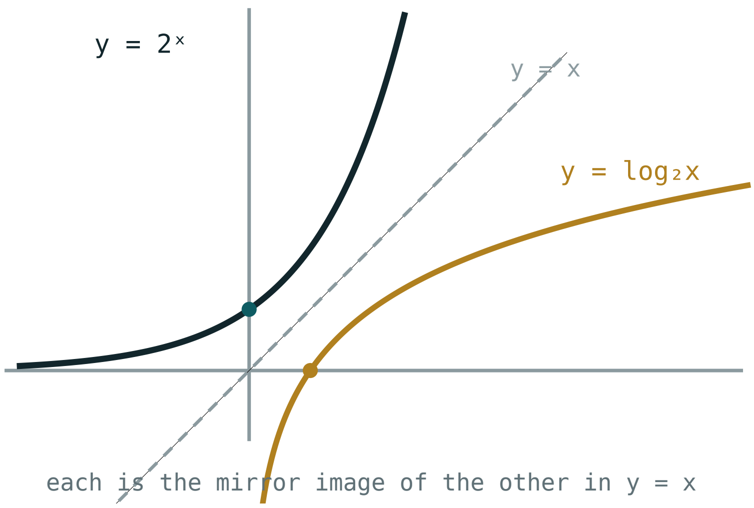
EXPRESSION	QUESTION	VALUE
$\log_2 32$	$2^? = 32$	5
$\log_3 1$	$3^? = 1$	0
$\log_7 7$	$7^? = 7$	1
$\log_4 2$	$4^? = 2$	$\frac{1}{2}$
$\log_2 \frac{1}{8}$	$2^? = \frac{1}{8}$	-3
$\lg 1000$	$10^? = 1000$	3

Two notations are standard: $\boldsymbol{\lg x}$ means $\log_{10} x$, and $\boldsymbol{\ln x}$ means $\log_e x$, where $e \approx 2.718$ is the base that makes calculus simplest.

$$\log_a 1 = 0 \cdot \log_a a = 1 \cdot a^{\log_a b} = b$$

The last is the **fundamental logarithmic identity**: raising a to the power that gives b gives b. It is the statement that the two operations undo each other.

The graph



Reflect $y = a^x$ in the line $y = x$ and the logarithm appears, with the axes' roles swapped.

PROPERTY	$Y = A^x$	$Y = LOG\_A X$
domain	$\mathbb{R}$	$x > 0$
range	$y > 0$	$\mathbb{R}$
passes through	$(0, 1)$	$(1, 0)$
asymptote	$y = 0$	$x = 0$
$a > 1$	increasing	increasing
$0 < a < 1$	decreasing	decreasing

Every row is the previous table with the two columns swapped — because reflecting in $y = x$ exchanges the axes.

Inverse, and why it matters

$$log\_a(a^x) = x \quad a\^(log\_a x) = x$$

These are the two halves of "$f^{-1}f = ff^{-1}$ = identity" from Grade 10 Quarter I.

WHAT LOGARITHMS ARE FOR

They take an unknown **out of an index**. $2^x = 10$ cannot be solved by matching bases — but taking log_2 of both sides gives $x = log_2 10 \approx 3.32$ immediately. That is the whole reason the function exists.

02 Worked examples

EXAMPLE 1 **Evaluate $\log_3 81$, $\log_5 \frac{1}{125}$ and $\log_8 2$.**

1. $3^4 = 81$

   4

2. $5^{(-3)} = \frac{1}{125}$

   -3

3. $8^{(1/3)} = 2$

   $\frac{1}{3}$

ANSWER

$4, -3, \frac{1}{3}$

EXAMPLE 2 **Solve $\log_2 x = 5$ and $\log_x 49 = 2$.**

1. $x = 2^5 = 32$

2. $x^2 = 49$

3. $x = 7$; the base must be positive and not 1.

   $x = -7$ rejected.

ANSWER

$x = 32; x = 7$

EXAMPLE 3 **Find the domain of $y = \log_3(x - 4)$.**

1. The argument must be positive.
2. $x - 4 > 0$
3. $x > 4$

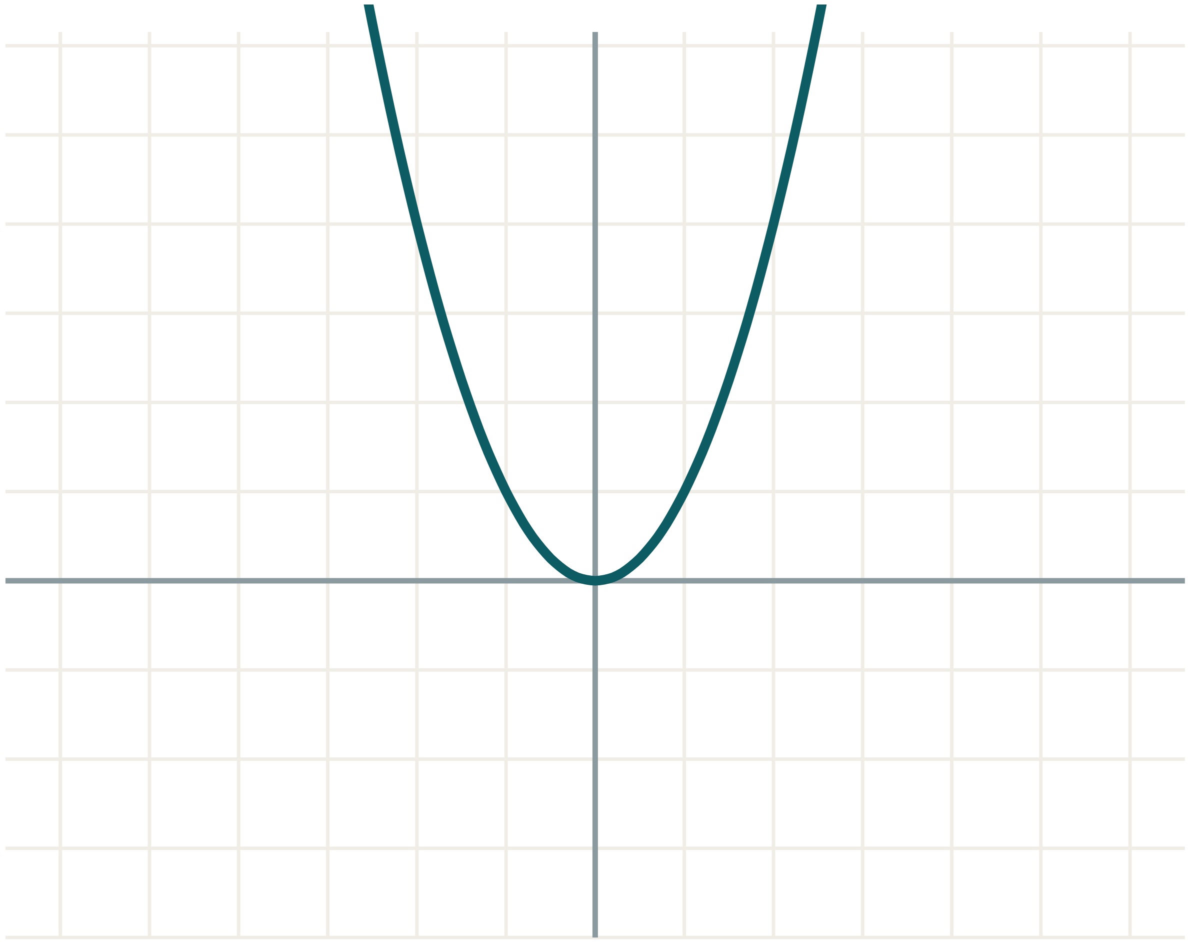
ANSWER

$x > 4$

03 Interactive model

Say each logarithm aloud as a question — “two to what power gives eight?”

The logarithmic curve



a **1** | b **0** | c **0** | equation `y = f(x)`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Logarithm	Logarifm	Логарифм
Base of a logarithm	Logarifm asosi	Основание логарифма
Argument of a logarithm	Logarifm argumenti	Аргумент логарифма
Common logarithm (lg)	O‘nli logarifm	Десятичный логарифм
Natural logarithm (ln)	Natural logarifm	Натуральный логарифм
Vertical asymptote	Vertikal asimptota	Вертикальная асимптота
Inverse function	Teskari funksiya	Обратная функция
Logarithmic identity	Logarifmik ayniyat	Основное логарифмическое тождество
Reflection in y = x	y = x ga nisbatan akslantirish	Симметрия относительно y = x

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$log_2 16$ is:

- 2
- 4
- 8
- 32

$log\_a 1$ is:

- 0
- 1
- a
- undefined

The domain of $y = log\_a x$ is:

- $\mathbb{R}$
- $x > 0$
- $x \geq 0$
- $x \neq 0$

$y = log\_a\ x$ passes through:

- $(0, 1)$
- $(1, 0)$
- $(0, 0)$
- $(1, 1)$

$a^{(log\_a\ 7)}$ equals:

- a
- 7
- $log\_a\ 7$
- 1

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\log_2 8$

ANSWER 3

2. $\log_3 9$

ANSWER 2

3. $\log_5 1$

ANSWER 0

4. $\log_7 7$

ANSWER 1

5. $\lg 100$

ANSWER 2

6. $\log_2 \dfrac{1}{4}$

ANSWER -2

7. $\log_9 3$

ANSWER $\dfrac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $log_3 81$

ANSWER 4

2. $log_5 \frac{1}{125}$

ANSWER -3

3. $log_8 2$

ANSWER $\frac{1}{3}$

4. Solve $log_2 x = 6$

ANSWER $x = 64$

5. Solve $log\_x\ 49 = 2$

ANSWER $x = 7$

6. Domain of $log_3(x - 4)$

ANSWER $x > 4$

7. $2\text{^}(log_2 9)$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $log_4 \frac{1}{32}$

ANSWER -2.5

2. $log\_(0.5)\ 8$

ANSWER -3

3. Solve $log\_x\ 8 = \frac{3}{2}$

ANSWER $x = 4$

4. Domain of $log_2(x^2 - 9)$

ANSWER $x < -3$ or $x > 3$

5. Domain of $log_5(6 - x) + log_5(x - 1)$

ANSWER $1 < x < 6$

6. Solve $3\hat{}(log_3 5 + 1)$

ANSWER 15

7. Explain why $log_2(-4)$ does not exist

ANSWER No power of 2 is negative

07 Homework

Homework — 5 tasks

Read every logarithm aloud as a question before answering it.

1. Evaluate $log_2 64$, $log_3 \frac{1}{27}$, $log_{16} 4$ and $lg\ 0.001$.
2. Solve $log_5 x = 3$ and $log\_x\ 64 = 3$.
3. Find the domain of $y = log_4(2x - 6)$ and of $y = log_2(x^2 - 4)$.
4. Sketch $y = log_2 x$ and $y = 2^x$ on one set of axes with the line $y = x$.

5. Explain in three sentences why the argument of a logarithm must be positive.